1. Figures 1 (a, b c, and d) show four popular biasing schemes typically used in analogue integrated circuits.

   (a) Explain the function of each of the circuits in Figure 1 (a, b, c and d) and derive expressions for the constant output parameter in each case, clearly indicating component design requirements and any approximations you have made. You may ignore bulk effects in the CMOS circuits.

Band Gap Reference "Diode"

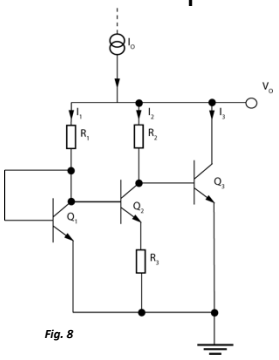


Fig. 8

Such device is very useful in applications where precision is critical such as high resolution ADCs

Though the Zener Diode can have a low temperature coefficient, the best "diode" for virtually zero temperature coefficient is the so called **"band-gap diode"**

It is a **2 terminal device**, not a single diode at all!

The algebra on the next pages will show that if V_O is set through the other components to be **= 1.283V**, then:

$$\frac{dV_o}{dT} \approx 0 \text{ !!}$$

Using the equation $I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right)$ and given that all the 3 BJTs are well matched and at the same temperature T, then from the circuit:

$$V_{BE1} = V_{BE2} + I_2 R_3$$

Note: all β >> 1

Then:

$$V_{BE1} - V_{BE2} = V_T \ln\left(\frac{I_1}{I_2}\right) = I_2 R_3$$

So:

$$V_O = V_{BE3} + \left(\frac{R_2}{R_3}\right) V_T \ln\left(\frac{I_1}{I_2}\right)$$

$$\frac{dV_O}{dT} = \frac{dV_{BE3}}{dT} + \frac{R_2}{R_3} V_T \frac{d}{dT} \ln\left(\frac{I_1}{I_2}\right)$$

13

Virtually zero T coefficient if

$$\frac{dV_{BE3}}{dT} = \left(\frac{V_T}{T}\right)\left(\frac{R_2}{R_3}\right)\ln\left(\frac{I_1}{I_2}\right)$$

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Buffered Voltage Source/Voltage Regulator

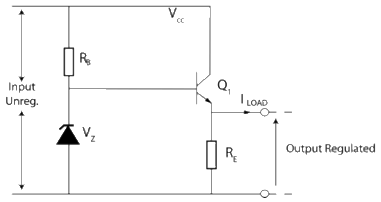


Fig. 6

$$V_O \gg (V_Z - V_{BE})$$

$$R_O \gg \frac{1}{gm}$$

R_B - **biases** the Zener (clearly the input must be greater than the Zener Breakdown voltage)

Q_1 – Common Collector, or emitter follower, providing **low output impedance** and close to **unity voltage gain**

R_E – Keeps Q_1 **conducting at all time**, even if the load current drops to zero

9

VGS multiplier:

V_{GS} Multiplier

Stolen from the V_{BE} multiplier

Circuit Analysis

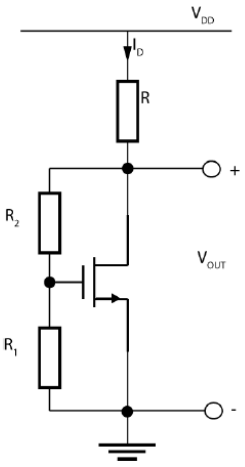


Fig. 16

$$V_{OUT} = V_{GS}\left[1 + \frac{R_2}{R_1}\right]$$

OR

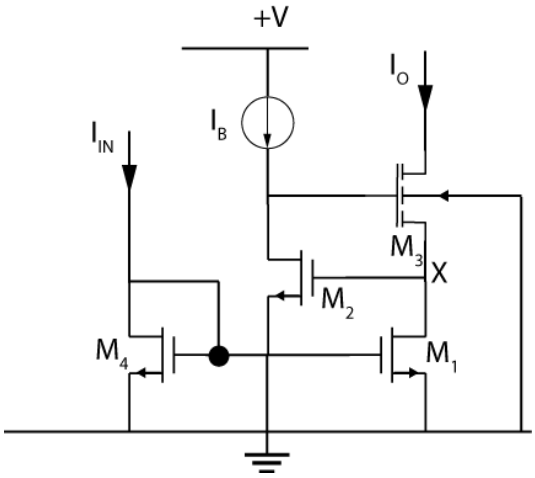
$$V_{OUT} = \left[1 + \frac{R_2}{R_1}\right]\left(V_T + \sqrt{\frac{V_{DD}}{\beta R}}\right)$$

Note: All R's can be replaced with active resistors

25

Regulated Cascode

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Circuit is a Regulated Cascode current-mirror. Circuit results in a very high output impedance.

Thumbs

M_4 cascodes m_2 and keeps v_{DS2} fairly constant giving an output resistance.

$$r_O » r_{DS2} g_{m4} r_{DS4}$$

(M_3 and I_B) form inverting amplifier and regulates voltage V_{DS3} so that output resistance is further increased by loop gain

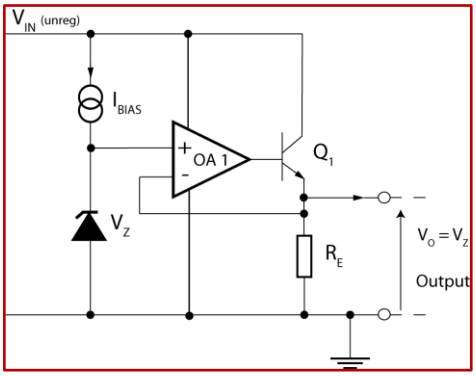
$$\frac{g_{m3}}{g_{O3}} = g_{m3} r_{DS3}$$

$$\therefore \quad R_{OUT} = \frac{v_O}{i_O} = g_{m4} g_{m3} r_{DS2} r_{DS4} r_{DS3}$$

$$» g_m^2 r_{DS}^3$$ very high

(b) Draw an improved version of the circuit Fig. 1a which also guarantees the voltage V_z is received by the output load and explain its operation.

[4]



- Clearly variations of the input voltage will cause **small variations** of Zener Voltage, though reduced approximately by the ratio R_Z / R_B

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- This is improved by driving the Zener with a **perfect DC current source** (see next part of this session)
- Also the input impedance and the dependence on the V_{BE} of Q_1 is reduced greatly by **using an Op-Amp with negative feedback**

Band Gap Reference "Diode"

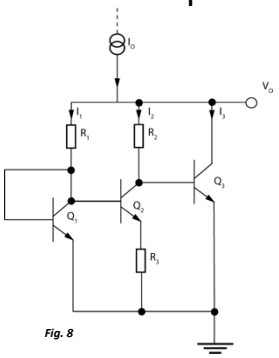


Fig. 8

Such device is very useful in applications where precision is critical such as high resolution ADCs

Though the Zener Diode can have a low temperature coefficient, the best "diode" for virtually zero temperature coefficient is the so called **"band-gap diode"**

It is a **2 terminal device**, not a single diode at all!

The algebra on the next pages will show that if V_O is set through the other components to be **= 1.283V**, then:

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$$\frac{dV_O}{dT} = \frac{dV_{BE3}}{dT} + \frac{R_2}{R_3} V_T \frac{d}{dT} \ln\left(\frac{I_1}{I_2}\right)$$

13

Virtually zero T coefficient if

$$\frac{dV_{BE3}}{dT} = \left(\frac{V_T}{T}\right)\left(\frac{R_2}{R_3}\right)\ln\left(\frac{I_1}{I_2}\right)$$

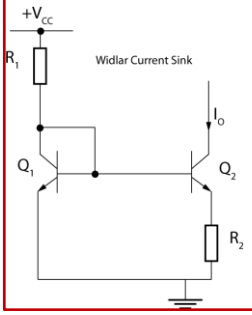
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2. (a) Figure. 2a shows a commonly used circuit for providing a stable current reference. The circuit has the following values I_O = 5μA, R_1 = 14.4KΩ, I_{IN} = 1mA, R_2 = 26.5KΩ. R2 is a diffused resistor with a temperature coefficient of +1500ppm/°C at T=300K. The output current sensitivity to the supply voltage V_{cc} is given by:

$$S_{V_{CC}}^{Io} = \frac{1}{1+\frac{I_O R_2}{V_T}} S_{V_{CC}}^{I_{IN}}$$

i) Derive the term S_{VCC}^{Iin} and thus the output current sensitivity.

EXAMPLE



$$I_O = \frac{\Delta V_{be}}{R_2} \quad \text{So} \quad I_O = V_T \frac{\ln\left[\frac{I_O}{I_{IN}}\right]}{R_2}$$

Can Show:

$$S_{V_{CC}}^{Io} = \frac{1}{1+\frac{I_O R_2}{V_T}} S_{V_{CC}}^{I_{IN}}$$

$$I_{IN} = \frac{(V_{CC} - V_{be})}{R_1} \therefore S_{V_{CC}}^{I_{IN}} = \frac{V_{CC}}{V_{CC} - V_{be}}$$

Since $V_{CC} >> V_{be}$ then $S_{V_{CC}}^{I_{IN}} = 1$

[4]

ii) Calculate the change in I_o if V_{CC} where to change by 20%.

A 1% change in V_{CC} causes a 1% change in I_{IN}

$$S_{V_{CC}}^{Io} = \frac{1}{1+\frac{I_O R_2}{V_T}}$$

EXAMPLE

I_O = 5μA, R_1 = 14.4KΩ, I_{IN} = 1mA, R_2 = 26.5KΩ

Thus $$S_{V_{CC}}^{Io} = \frac{1}{1+\frac{5\times10^{-6}\times26.5\times10^{3}}{25\times10^{-3}}}$$

$$S_{V_{CC}}^{I_o} = 0.16$$

Therefore a 20% change in V_{CC} will cause I0 to change by 3.2%.

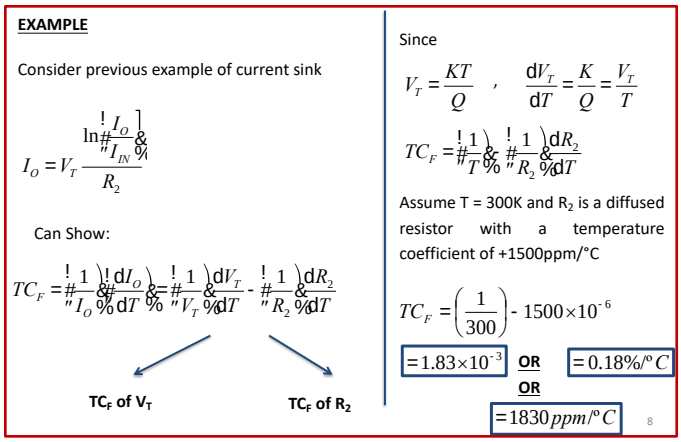
[4]

iii) The fractional temperature coefficient of the circuit is given by:

$$TC_F = \left(\frac{1}{I_O}\right)\left(\frac{\delta I_O}{\delta T}\right) = \left(\frac{1}{V_T}\right)\frac{\delta V_T}{\delta T} - \left(\frac{1}{R_2}\right)\frac{\delta R_2}{\delta T}$$

Calculate the value of TC_F stating assumptions where necessary.

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[4]

(b) Figure 2b shows an improvement to the circuit in Figure 2a.

i) Describe two improvements this circuit has for generating a constant bias.

1) Self biased current reference making it supply independent.
2) Start-up circuit to always ensure correct operating point.

[4]

ii) Calculate the fractional temperature coefficient for this circut at room temperature, given that *R* is a polysilicon resistor with a temperature coefficient of 1500 ppm/°C, the temperature coefficient of V_{BE} to be *-2.5mV/°C* both at T=300K. State all assumptions.

[4]

Equation for output current:

$$I_O = \frac{(V_{be1} - V_{be2})}{R} = \frac{V_T \ln\left[\frac{I_{IN}}{I_O}\right]}{R} = \frac{V_T}{R}\ln\left[\frac{I_{IN}}{I_{S1}} \times \frac{I_{S2}}{I_O}\right]$$

If we self bias and force $I_{IN}/I_O = 1$ with top current mirror, then ln(1) = 0
Require $I_{IN}/I_O > 1$
In Figure 2.1, $I_{IN}/I_O = n = 2$
So, assuming matched devices

$$I_O = \frac{V_T}{R}\ln[2]$$

Fig 2.1 is a Improved TCF using VT Referenced Current Source/Sink

$$TC_F = \left(\frac{1}{V_T}\frac{dV_T}{dT}\right) - \left(\frac{1}{R}\frac{dR}{dT}\right)$$

$$TC_F = \frac{1}{T} - \left(\frac{1}{R}\frac{dR}{dT}\right)$$

$$= \left(\frac{1}{300} - 1500x10^{-6}\right) = 1833\,ppm/^oC$$

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3. (a) Figure 2 shows a two-stage CMOS op-amp. Estimate the low-frequency differential voltage gain and gain-bandwidth product of the amplifier. Aspect ratios of all devices are shown on the circuit. Assume all bulk effects are negligible. Device model parameters are given below. What is the main advantage and disadvantage of a single-stage compared to a two-stage op-amp ?

$$Av1 = -\,gm2/(go2+go4)$$

$$(go2+go4) = I_{D2}(\lambda_N+\lambda_P) = 5\times10^{-6}\times0.05 = 2.5\times10^{-7}\,\mathrm{W}^{1}$$

$$gm2 = 2\sqrt{\beta_2 I_D}$$

$$\beta_2 = \frac{K_N}{2}\left(\frac{W}{L}\right)_2 = 7.5\times10^{-5}\,A/V$$

$$gm2 = 3.87\times10^{-5}\,S$$

$$Av1 = -\,154.9$$

$$Av2 = -\,gm6/(go7+go6)$$

$$(go7+go6) = I_{D6}(\lambda_N+\lambda_P) = 20\times10^{-6}\times0.05 = 10\times10^{-7}\,\mathrm{W}^{1}$$

$$gm6 = 2\sqrt{\beta_6 I_{D6}}$$

$$\beta_6 = \frac{K_N}{2}\left(\frac{W}{L}\right)_6 = 1.6\times10^{-4}\,A/V$$

$$gm6 = 1.13\times10^{-4}\,S$$

$$Av2 = -\,113$$

$$A_{total} = Av1\times Av2 = 17503$$

$$G.Bp = gm2/2\pi C_c = 12.32\,MHz$$

[8]

In a 2-stage opamp the load contributes to the 2nd pole hence reducing load increases stability.
With a single-stage, the load forms the dominant pole hence reducing the load increases bandwidth.

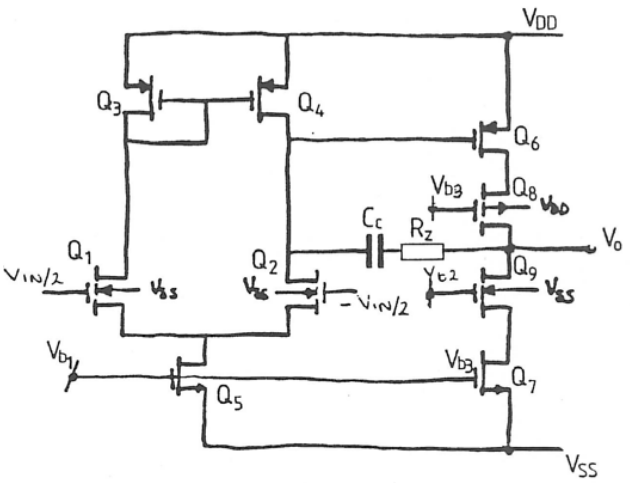
[2]

[10]

(b) Show how the output transistors can be cascoded to increase the voltage gain. State two advantages and two disadvantages of a cascoded output stage.

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2. CASCODED OUTPUT



[6]

Advantages:
Cascoding output stage increases the output impedance by:
Go=[go6(go8/gm8) +go7(go9/gm9)]
Since Gain of output stage=gm6/[Go]=A_{V2}
Therefore we increase the output impedance and the gain. [2]

Disadvantages:
We have reduced output voltage headroom due to the 4 transistors which need to be in saturation.
Additional bias circuitry is needed to bias the cascade transistors. [2]

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4. a) Under which operating conditions does the MOSFET of Figure 3.1 realise a linear floating resistor between terminals *A* and *B* ? Explain why the bulk is not connected to the source.

NON-SATURATION REGION

Recall: $V_{DS} < V_{GS} - V_T$

$$I_D = \left(\frac{KW}{L}\right)\left[\left(V_{GS} - V_T\right)V_{DS} - \frac{V_{DS}^{\ 2}}{2}\right]\left(1 + \lambda V_{DS}\right)$$

Assuming $V_{DS} << V_{GS} - V_T$ And so $\lambda V_{DS} << 1$

Then $I_D = \left(\frac{KW}{L}\right)\left(V_{GS} - V_T\right)V_{DS}$ [4]

The Bulk is not connected to the source due to void VBS from becoming a forward biased diode.

(b) Discuss three sources of non-linearity in the single MOSFET resistor realisation of Figure 4.1 and suggest one suitable circuit design to help to eliminate one or more of these non-linear terms. Show all necessary circuit analysis to confirm your design.

[6]

Problems with Simple Floating Active Resistors

1. Linearity limited due to **V_{BS} changing V_T for negative V_{DS}**

   Recall: $V_T = V_{TO} + \gamma\left[\left(\sqrt{-V_{BS} + 2\Phi F}\right) - \sqrt{2\Phi F}\right]$

   V_T is very nonlinearly related to V_{BS}

   Y = bulk threshold potential, ΦF = Fermi level potential

2. Linearity limited due to V_{DS} approaching ($V_{GS} - V_T$) hence **saturation region** for large positive V_{DS}

3. For **large values of V_{DS}** the $V_{DS}^2/2$ term comes in making the resistor quite nonlinear

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Consider the following **PARALLEL CIRCUIT**

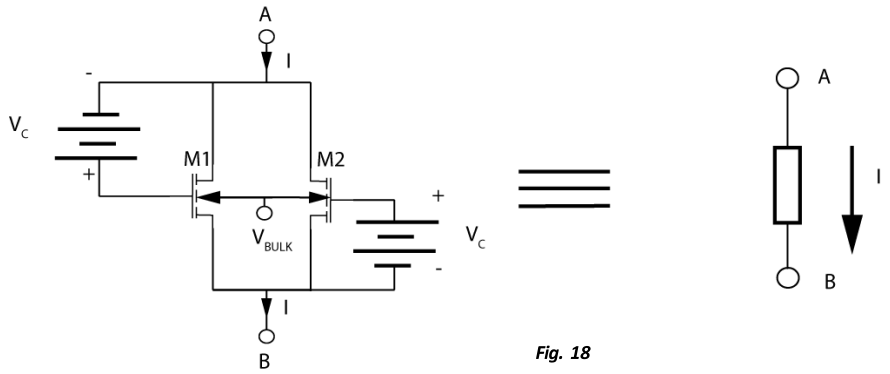


Fig. 18

Assuming both devices identical and non-saturated then:

$$I_1 = \left(\frac{KW}{L}\right)\left[(V_{DS} + V_C - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right] \qquad I_2 = \left(\frac{KW}{L}\right)\left[(V_C - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

30

Since $I = I_1 + I_2$

Then $$\frac{V_{DS}}{I} = R_{AB} = \frac{1}{\left(\frac{2KW}{L}\right)[V_C - V_T]}$$ ⇒ **LINEAR**

Tuning Range determined by V_C which is independant of the signal unlike the single MOSFET. Note the **validity of the above only** for $V_{DS} < (V_C - V_T)$, hence **large V_C**, larger V_{DS} and so wider range of linearity. Results in about **twice linear range of previous circuit.**

(c) Show how the two transistor differential integrator shown in Figure 4.2 can be modified to create a double differential integrator. Sketch the new circuit and derive the time constant. [4]

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Double Differential MOS Active Resistor (2)

Circuit Analysis

Principal Equations

$$I_{D1} = 2b\left[(V_{C1} - V - V_T)(V_1 - V) - \frac{1}{2}(V_1 - V)^2\right]$$

$$I_{D2} = 2b\left[(V_{C2} - V - V_T)(V_1 - V) - \frac{1}{2}(V_1 - V)^2\right]$$

$$I_{D3} = 2b\left[(V_{C2} - V - V_T)(V_2 - V) - \frac{1}{2}(V_2 - V)^2\right]$$

$$I_{D4} = 2b\left[(V_{C1} - V - V_T)(V_2 - V_1) - \frac{1}{2}(V_2 - V)^2\right]$$

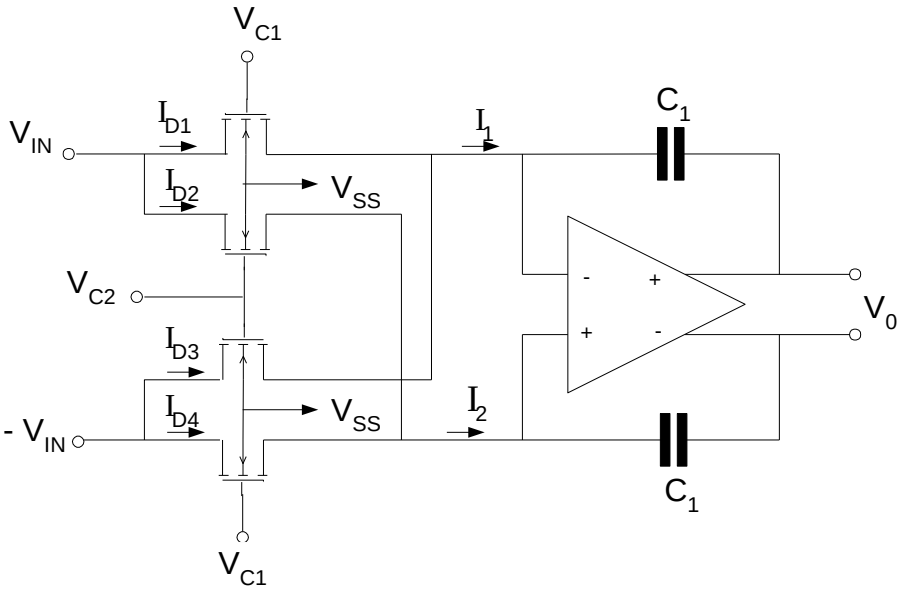
By expanding each expression out, it can be shown that

R(transresistance) =

$$\frac{V_1 - V_2}{I_1 - I_2} = \frac{1}{2b(V_{C1} - V_{C2})}$$

Hence R has a very wide tuning range and excellent linearity, independent of V_T

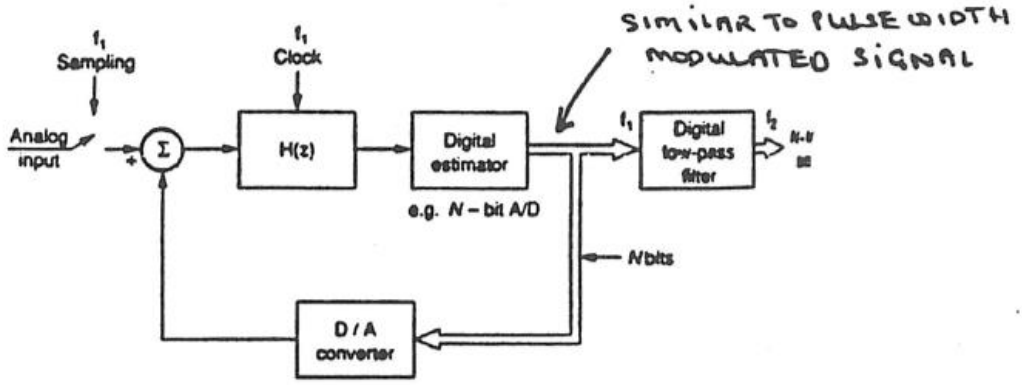
Note: We now must ensure $V_1, V_2 < (V_{C1} - V_T), (V_{C2} - V_T)$
i.e. $V_{DS} < V_{GS} - V_T$



(d) The integrator in Figure 4 is to be used in a biomedical application as part of an ECG patch to measure rhythms of the heart. The output voltage needs to be quantised with low noise and high dynamic range. Sketch a suitable ADC which suits this purpose and explain it's operation.

Explain sigma delta operation (bookwork).

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Typical oversampled Σ-Δ (modulator) converter architecture .

[6]

Explain how the conversion works (bookwork)

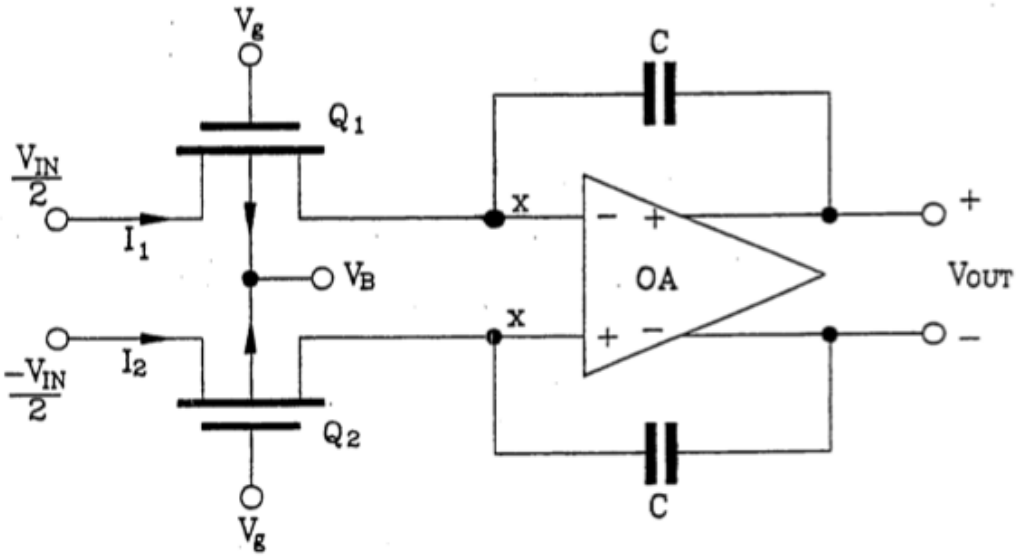


Figure 4.

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5. Figure 5 shows the basic design of an analogue sampled-data precision integrator.

(a) Derive an expression for the transfer function of the integrator. Assume that the integrator is driven by non-overlapping clocks and that the switches are ideal.

Derivation from notes

Termed "lossy" since only acts as an integrator at high frequency f > f_o

Verify above transfer function?

SOLUTION

$\Phi_1 \quad I_{av} = \frac{C_1 V_{IN}}{T}$

$\Phi_2 \quad \frac{-C_1 V_{IN}}{T} = V_O\left[j\omega C_2 + \frac{C_3}{T}\right]$

Rearranging yields

$$\frac{-C_1 V_{IN}}{C_3} = V_O\left[\frac{j\omega C_2 T}{C_3} + 1\right]$$

$$\frac{V_O}{V_{IN}} = -\frac{C_1}{C_3}\frac{1}{\left[1 + \frac{j\omega C_2 T}{C_3}\right]}$$

$$= -\frac{C_1}{C_3}\frac{1}{\left[1 + \frac{jf}{f_O}\right]}$$

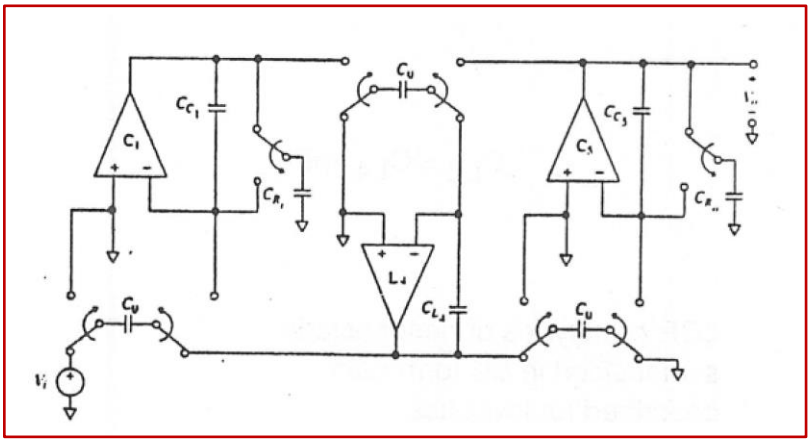
Where $f_O = \frac{1}{2\pi}\frac{C_3}{C_2} f_C$

[10]

(b) (i) Sketch the basic design of a 3rd-order Chebyshev low pass switched-capacitor ladder filter.

The filter is to have a cut-off frequency of 3kHz. Assume a clocking frequency of 200 kHz. The values of integration capacity for the capacitor based sections are 4.89pF and inductive section is 3.31pF. All other switched capacitors are 1pF.

(i) 3rd-order Chebyshev low pass switched-capacitor ladder filter.



[5]

General transformation rules for ladder prototypes:

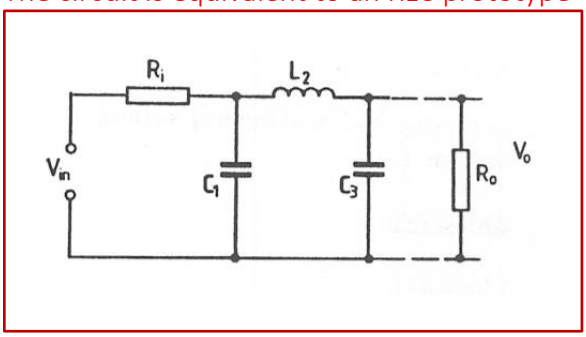
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Inductor: $\frac{f_c L_2}{R_s} = \frac{C_{L2}}{C_u}$

Capacitor: $\frac{C_{c3}}{C_u} = f_c R_s C_3$

Resistor Rs=dummy scalar.

The circuit is equivalent to an RLC prototype



(ii) From the circuit, estimate normalised passive component values for the original double-terminated LC prototype of the filter. All values should be normalised to 1 rad/s. You may assume that the clocking frequency is so high that the integrators can be assumed lossless.

For switched capacitor equivalent:

[5]

ii. For switched capacitor equivalent:

C_{c1}=Cc3=5.08 pF

C_{L2}=3.49 pF.

C_u=1pF

Assume scaling Rs=Ri=Ro=1Ω

Therefore

L2= CL2/fc= $3.49/100\text{x}10^3=3.49\text{x}10^{-5}$ H

Normalised 1rad/sec we multiply by $2\pi f_0$

L2=$3.49\ \text{x}10^{-5}$ x 2π x 5 KHz =1.096 H

C1=C3=Cc3/fc = $5.08/100\text{x}10^3 = 5.08\text{x}10^{-5}$ f

Normalised value C1=C3=1.596

[10]

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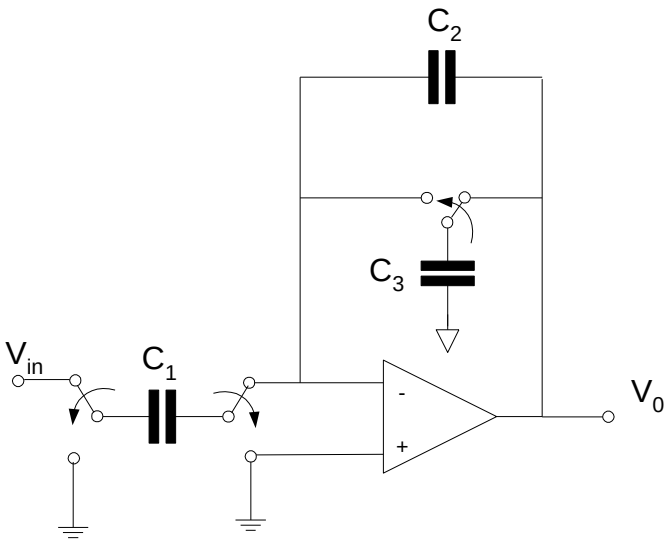
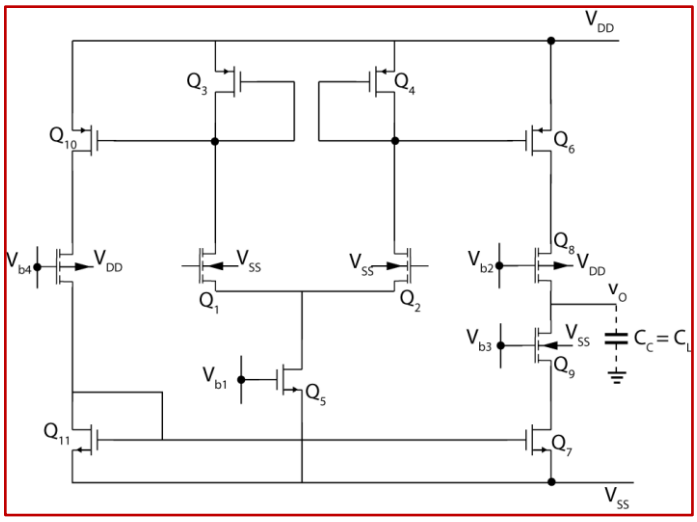


Figure 5.

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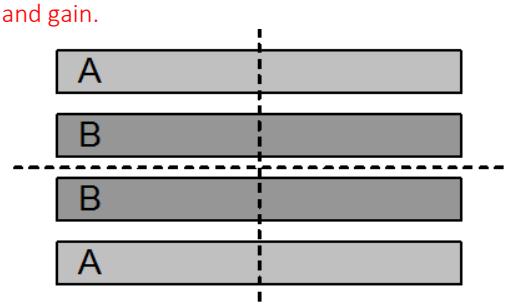
6. (a) Sketch the circuit of single-stage differential to single ended output high frequency operational amplifier that does not require miller compensation.



[5]

(b) When matching transistors special care is taken with the layout. Discuss what is meant by common centroiding when matching two transistors.

Mismatches like λ (the Vds difference), threshold voltage VT and W/L may cause errors in offset and gain. A proper layout may help reduce the effect of such mismatches as well as gradients. Common centroid design splits the transistors over a common centroid or axis of symmetry. For example, the elements A and B in an array may be inter-digitated as in figure below. The input differential pair transistors in the Op-amp are the critical ones for the layout as they affect the offset and gain.



[5]

(c) For fully differential output op-amps explain what is meant by common mode feedback using a diagram where necessary.

If an amplifier has a differential output with high gain, the voltages at the outputs are pretty well ill defined and will drift all over the place, usually taking the amplifier out of its high gain region.

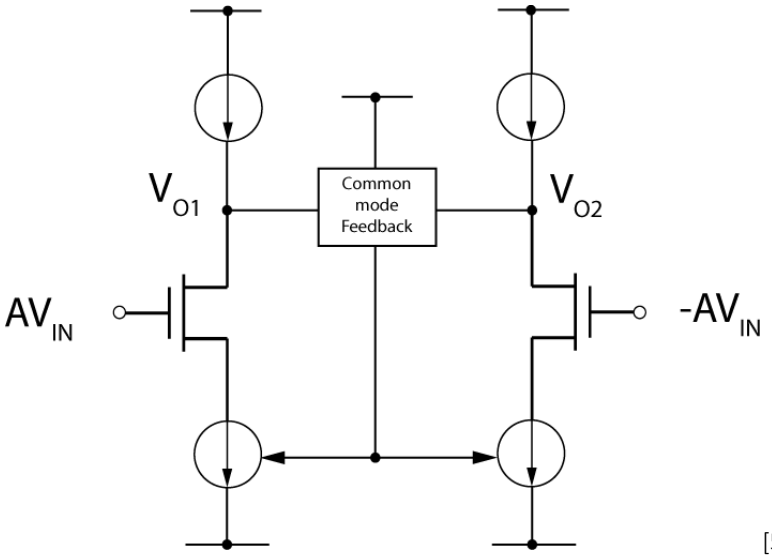
We therefore need to sample an output which is "common" to both and stabilise this voltage.

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The technique is known as common-mode feedback and we will describe it in slightly more detail when we come on to differential in, differential out, CMOS op-amp architectures.

We will show here how the technique can be used to improve the CMRR of the simple differential amplifier.

[10]



[5]

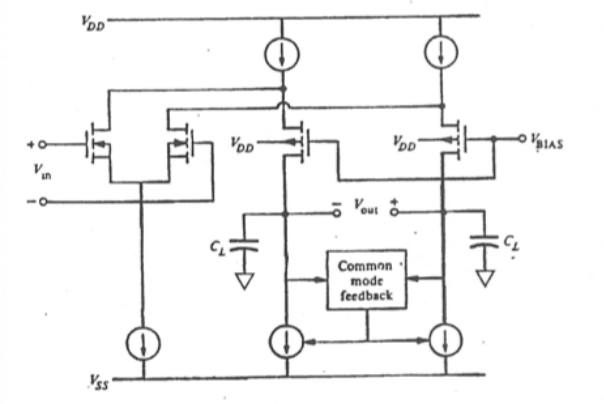
[5]

(d) Sketch the circuit of a differential operational amplifier that has a single dominant pole and uses common mode feedback to stabilise the output DC voltage.

[5]

In the folded cascode we are unstacking the conventional cascode and spreading it out. The AC current path is folded and this allow a reduction in power supply. The conventional cascode requires a 2-stage architecture and since the impedance at (x) is high requires internal compensation. The folded cascode can be used as a single stage architecture, node x is low impedance so the only high impedance node will be at the output.
Gain of the folded cascode Av=gm1/Go.

[4]



[6]

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